

Physiological Biometrics:

Fingerprint scan

- Capturing a detailed map of the friction ridges and valleys on your fingertip.
- Instead of taking a flat photograph, it uses specialized sensors to measure physical, electrical, or acoustic differences.
- The system then translates this unique pattern into a secure, digital code

There are three main types of sensors used in modern fingerprint scanning devices:

1. **Capacitive Scanners:** These are the most common (found on standard laptop and phone power buttons). The sensor contains thousands of microscopic capacitors that store electrical charges. When you place your finger on the sensor, ridges touch the surface while valleys create tiny air gaps. This variation in electrical contact maps the print
2. **Optical Scanners:** Often used for physical attendance systems or entry-level in-display readers, these scanners shine a light on your finger and take a high-contrast digital picture of the ridge-and-valley layout.

3. Ultrasonic Scanners:

- Found in premium smartphones, this technology uses high-frequency sound waves.
- The sound waves pass through the outer layers of the skin and bounce back, creating a highly accurate 3D map.
- These are incredibly secure and can even scan through water, dirt, or grease.



Components of finger scan system





Image Acquisition

- The surface on which the finger is placed is called a **platen**, also referred to simply as a **scanner**.
- Platens can be made of various materials, **including glass, plastic, silicon, and polymer**.
- Depending on the type of finger-scan technology used, areas of contact between the fingerprint and the platen are measured through **chip-based cameras, through ultrasonic imaging, or through changes in capacitive fields generated by the finger**.
- These measurements are converted into **digital code**, at which point they can be processed by the finger-scan system.

Image Acquisition

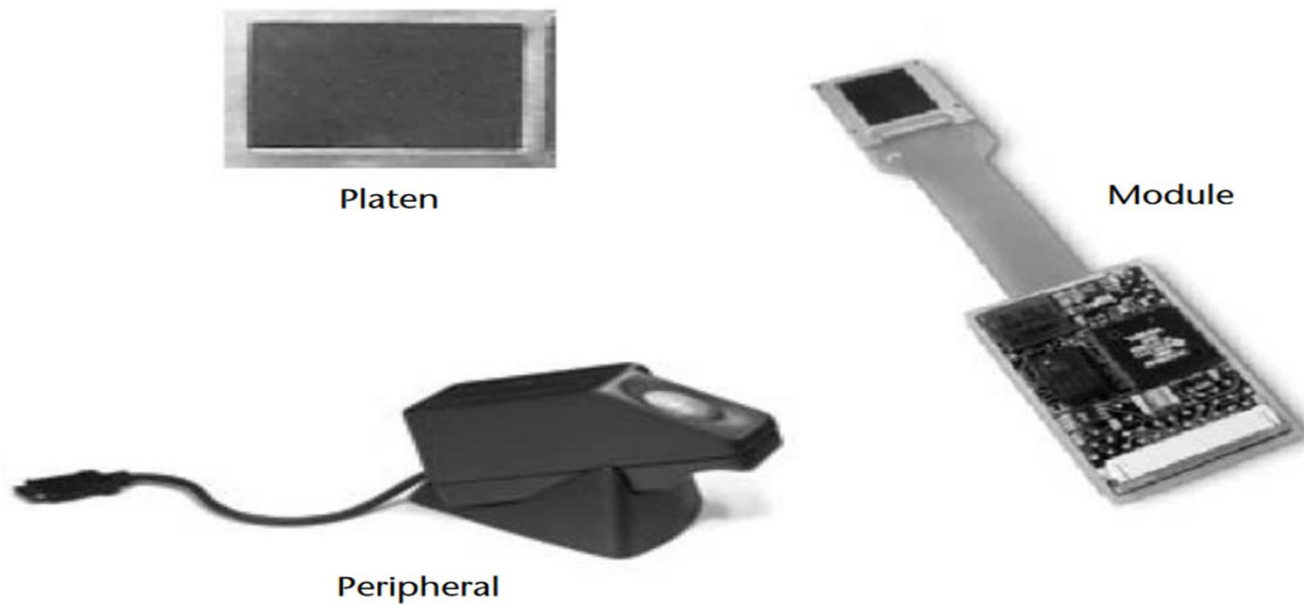


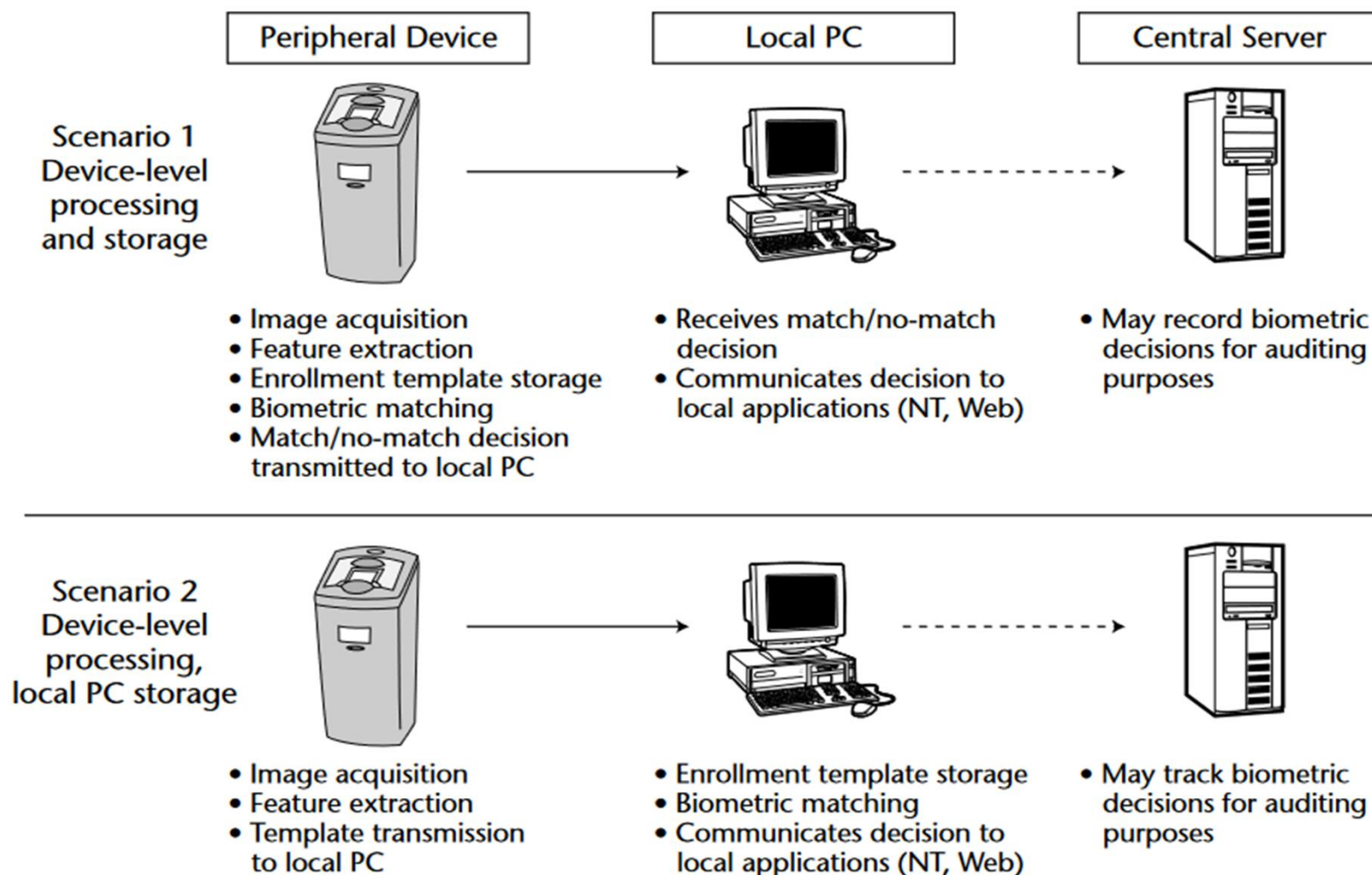
Figure 4.2 Components of a finger-scan system.



Image Acquisition

- A platen is one piece of a finger-scan module, the basic building block of a peripheral or standalone finger-scan device.
- A module normally contains a platen attached to a small printed circuit board, along with a standard connector that allows digitized information to be transmitted to the peripheral or standalone device.

How finger scan Technology works



How finger scan Technology works

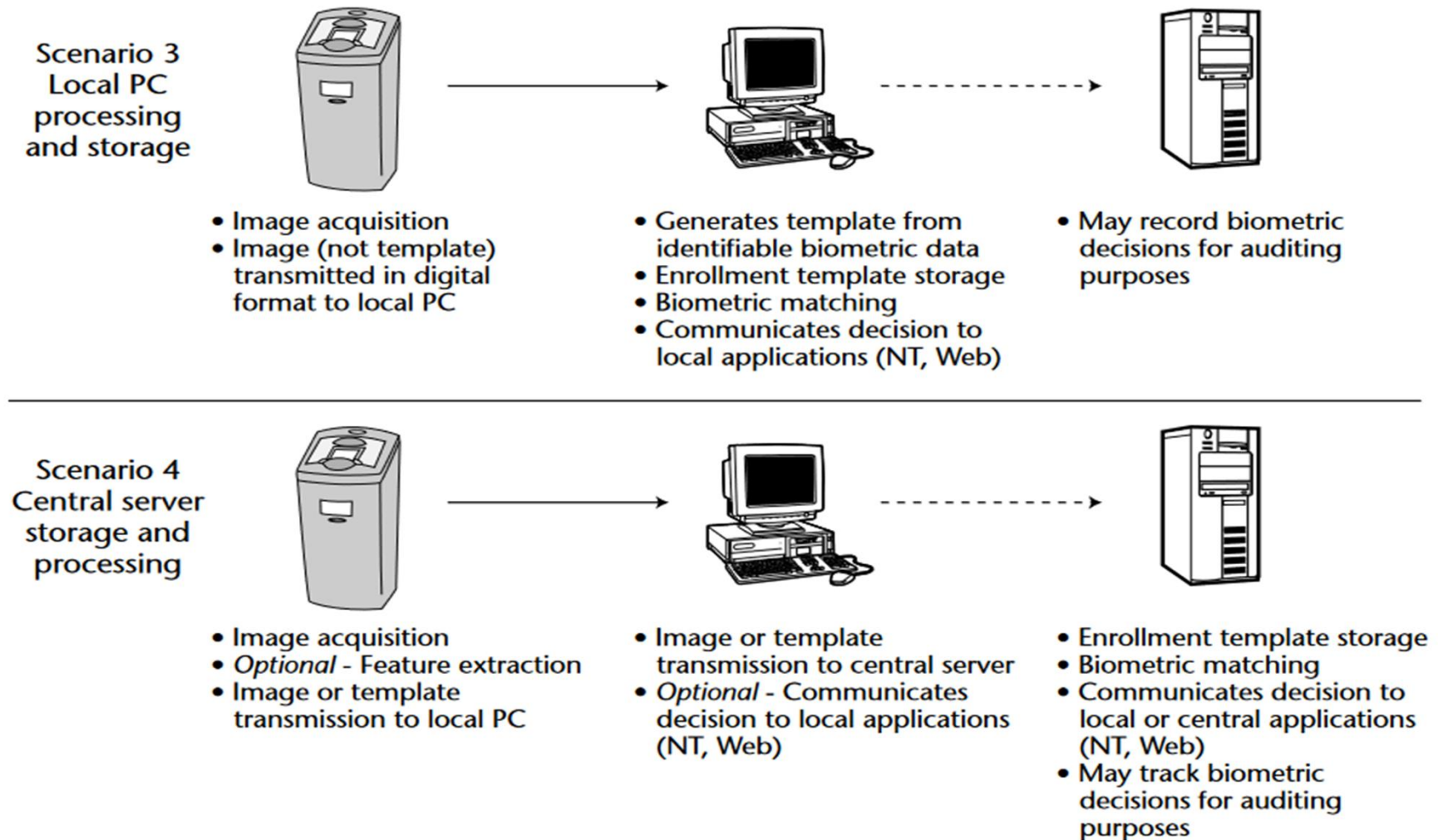


Figure 4.3 Schematic—data storage and processing in finger-scan systems.



Image Processing

- Once a high-quality image is acquired, it must be **converted to a usable format**.
- Image processing subroutines eliminate gray areas from the image by converting the fingerprint image's gray pixels to white and black, depending on their pitch.
- The ridges are then thinned from approximately 5 to 8 pixels in width down to a single pixel, for precise location of features.



Image Processing



Raw Image



Ridge Isolation



Thinned Ridges

Figure 4.5 Steps involved in image processing.

Types of Minutiae

- There are many types of minutiae
 - Ridge endings (the point at which a ridge ends)
 - Bifurcations (the point at which one ridge divides into two).



Figure 4.7 Finger-scan minutiae.



Template Creation

- Information used when mapping minutiae can include the location and angle of a minutia point, the type and quality of minutiae, and the distance and position of minutiae relative to the core.
- Finger-scan images will normally have distortions and false minutiae that must be filtered out before template creation.
- For example, anomalies caused by scars, sweat, or dirt can appear as minutiae.
- A large percentage of false minutiae are discarded in this process, ensuring that the template generated for enrollment or verification is an accurate reflection of the biometric data



Template Creation

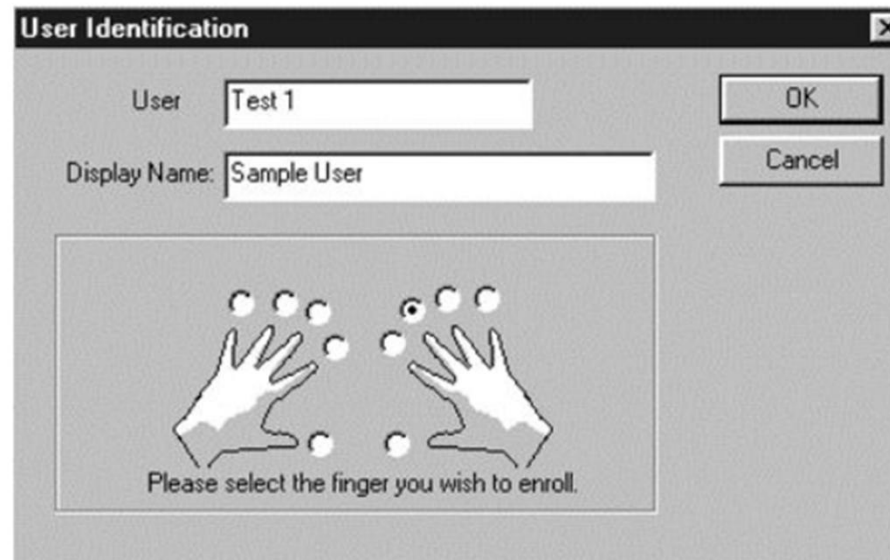


Figure 4.6 Finger-scan enrollment prompts.

Noise Filtering /Image Enhancement Algorithms

Algorithm	Purpose
Median Filter	Removes salt-and-pepper noise while preserving edges.
Gaussian Filter	Smooths Gaussian noise and reduces image variations.
Wiener Filter	Removes blur and noise adaptively.
Bilateral Filter	Reduces noise while preserving ridge edges.
Histogram Equalization	Improves image contrast.
CLAHE (Contrast Limited Adaptive Histogram Equalization)	Enhances local contrast without amplifying noise excessively.
Gabor Filter (<i>Most Common</i>)	Enhances fingerprint ridges based on local ridge orientation and frequency.
FFT-based Enhancement	Enhances periodic ridge structures using frequency-domain analysis.
Anisotropic Diffusion Filter	Smooths homogeneous regions while preserving ridge boundaries.

Ridge Thinning

- Converts ridges into one-pixel-wide skeletons.
- Algorithms
 - Zhang–Suen Thinning Algorithm (*Most Popular*)
 - Guo–Hall Thinning
 - Morphological Skeletonization

Template Generation Algorithms

- The fingerprint template is a compact representation rather than the full image.
- The template stores features such as:
 - Ridge ending coordinates (x, y)
 - Bifurcation coordinates
 - Orientation angle
 - Ridge quality score
 - Minutiae type
- Egs. Of minutiae extraction algorithms
 - Crossing Number (CN) Algorithm
 - Run-Length Encoding

Template Matching Algorithms

- After template generation, matching algorithms compare two templates.

Common methods are,

- Minutiae Matching
- Bozorth3 Algorithm (NIST)
- Graph Matching
- Hough Transform
- Euclidean Distance
- Cosine Similarity (for deep learning embeddings)

Security Threats and Vulnerabilities

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Stage	Threat	Vulnerability	Possible Impact
Fingerprint Sensor	Spoofing Attack	Fake fingerprints made from silicone, gelatin, latex, or 3D prints	Unauthorized access
Sensor	Replay Attack	Replaying previously captured fingerprint images	Bypass authentication
Preprocessing	Adversarial Image Attack	Maliciously modified fingerprint images	Incorrect feature extraction
Feature Extraction	Feature Manipulation	Altering minutiae points	False Match or False Reject

Communication Channel	Man-in-the-Middle (MITM)	Intercepting fingerprint data	Data theft or modification
Template Database	Template Theft	Stealing stored fingerprint templates	Identity theft
Template Database	Template Modification	Inserting or changing templates	Unauthorized enrollment
Matcher	Hill-Climbing Attack	Repeatedly modifying inputs until accepted	System compromise
Decision Module	Decision Override	Modifying authentication results	Unauthorized access
Entire System	Insider Attack	Administrator misuse	Data leakage

Countermeasures: spoofing attack

- Liveness Detection
- Multi-spectral fingerprint sensing
- Pulse detection
- Sweat pore detection
- Temperature sensing

- Can you identify **Countermeasures of other identified threats such as Replay Attack, Adversarial Image Attack, Feature Manipulation**
- **, Template Theft, Template Modification and insider attack**
- **(Refer Previous slide table)**